

The dynamic sector

Kinetic energy as a ξ -fractal fraction of the static ($\tilde{T} \cdot m = 1$): problem statement, narrowing, and the structural lead

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Abstract

Through the duality $\tilde{T} \cdot m = 1$, FFGFT separates two energy types: *static* (rest mass = topologically frozen winding, the crystallographic Z_3 core) and *kinetic* (free; for massless states the only form, $E = hf$). The static sector is derived and computed (Z_3 mass operator + mass circle; Doc. 285, BD12-XCORE). The dynamic sector is *not*: the derived rule that computes the kinetic part, via the duality, as a ξ -fractal fraction of the static one is missing. This document is a *problem-statement document* – it records what is fixed (the duality places the window in the kinetic sector), the single existing data point (neutrino, $m_\nu \sim \xi^2/2 m_e$, as an ansatz), the narrowing of the ξ -power to $p = 2$, and the structural lead $m_{\text{window}} = \xi \cdot \omega$. What remains to be derived is marked explicitly as open. Prefixed is the fixed, reversible Hilbert-space foundation (Doc. 282/270/230) the sector sits on; its actual aim is to find, out of the Hilbert space, the complementary description that maps via $D_f = 3 - \xi$ and ξ onto SI units.

1 Where the question comes from

In Doc. 255, $\tilde{T} \cdot m = 1$ enforces two energy types. The rest mass is *static*: a topologically stable winding number on T^4 , fixed in the compact picture by the Z_3 mass operator ($r = \sqrt{2}$, $\theta = 2/9$). The kinetic energy is *free*; for a massless particle it is the only form ($E = hf$). "Mass is frozen kinetic energy."

Doc. 285 shows that the empty recursion windows ($3'$, $3''$, ...) carry no *new* masses but structure; and that a *dynamics* filling a window supplies the *kinetic* type – a filled window is photon-/neutrino-like, not a massive triple. There the gap was named: the *derived* rule computing the kinetic part, via the duality, as a ξ -fractal fraction of the static one is missing. This document takes up exactly that gap. It is the "further work" on the dynamic sector announced in Doc. 285.

2 The fixed foundation: the reversible Hilbert-space basis

Before developing the dynamic sector, let the fixed foundation it sits on be recorded. The FFGFT QM translation (Doc. 282) is closed in three pieces:

1. **The unambiguous Hilbert map** (Type III, Doc. 270): $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$, states = fiber-valued fields, QM operation = unitary parallel transport, entanglement = holonomy around cycles (Doc. 230), three masses $\leftrightarrow (M, r, \theta)$. A *bijection* – fully reversible.
2. **The time-free mass form**: mass sits on a separate tensor factor; CHSH = $2\sqrt{2}$ stays identical with and without the $\mathbb{C}^3$ fiber to $4 \cdot 10^{-16}$ – no static QM result changes. Time is *not computed* here; the reduced dynamics is time-local, unitary, Markovian.
3. **The time form**: adds exactly *one* effect – non-Markovian memory in the reduced dynamics (backflow $0 \rightarrow 0.53$, CP-divisibility violated) – which itself still lies within QM: a traced-out unitary bath reproduces it (Doc. 282/283). At the full level, therefore, unitary/reversible.

Reversibility is precise *per type* (Doc. 270): Type III (Hilbert) fully reversible; Type I (mass circle $\rightarrow$ time, "time taken into the space") *almost* reversible – lossless in mode content, only the winding number stays ambiguous; Type II (spatial projection $D3 \rightarrow D0$) *not* reversible. The only genuine irreversibility is the spatial projection – the measurement / decompactification (Doc. 270/284) –, not the time-inclusion. The non-Markovian memory is not a break of reversibility but the reduced view.

Aim of the dynamic sector. On this fixed, reversible foundation, the actual task is the *Hilbert-space extension and its translation: another description out of the Hilbert space* – the complementary, decompactified/projected side that maps computationally, via $D_f = 3 - \xi$ and ξ , onto SI units. The SI map is exactly the measurement/projection side (Type I/II), and D_f and ξ are the bridge quantities (Doc. 270/249/133). The dynamic sector is thus not a departure from the foundation but its unrolling into measurable units – and the sought kinetic ξ -fraction is one building block of exactly that unrolling.

.3 What is fixed

The duality fixes the *placement*, not the size. With $\tilde{T} = 1/\max(m, \omega)$ (Doc. 067, 080), an almost-massless window state has $\omega \gg m$, hence $\tilde{T} = 1/\omega$: the state sits in the *kinetic* sector. That is secure. So is the conceptual separation:

	static	kinetic
form	$E_0 = mc^2$ (rest)	$E = hf$ (free)
seat	Z_3 core (winding)	window, when a dynamics runs
time	$\tilde{T} = 1/m$	$\tilde{T} = 1/\omega$
status	derived (Doc. 285)	<i>open</i> (this document)

A distinction between two kinetic quantities matters: the ordinary kinetic energy of the electron, $(\gamma - 1)mc^2$ (Doc. 080), is *not* ξ -suppressed. The ξ -fractal fraction sought here is the *window rest-mass* of an otherwise massless state – not the kinetic energy of a massive one.

.4 The single data point: the neutrino

Doc. 047 treats the neutrino as a “near-massless photon”: $E^2 = (pc)^2 + (\sqrt{\xi^2/2} mc^2)^2$, with $v_\nu = c(1 - \xi^2/2) \approx 0.9999999911c$ and $m_\nu \sim \frac{\xi^2}{2} m_e$. This is the only existing value for a window rest-mass as a ξ -power of a core mass. It is, however, explicitly an *ansatz* (from the velocity/photon analogy), not *derived* from the duality. Doc. 204 additionally supplies a ξ -correction to the kinetic energy ($+0.69\xi_0$). There are no further data points; in particular, a single value is not enough for the Δm^2 spectrum.

.5 Narrowing the ξ -power: $p = 2$

With the ansatz $m_{\text{window}} = C \cdot \xi^p \cdot m_{\text{ref}}$, tested against the bounds (reproducing the $\xi^2/2$ value; the photon limit; dimensional consistency; an oscillation floor $\gtrsim \sqrt{\Delta m_{\text{atm}}^2} \approx 49$ meV; a direct kinematic bound $m_\nu < 0.45$ eV (KATRIN 2025, 90 % CL, model-independent)), only $p = 2$ is compatible with the meV scale: $p = 1$ gives $\mathcal{O}(10\text{--}70)$ eV (far above the bound), $p \geq 3$ falls below every anchor. So $p = 2$ is the *unique* order-of-magnitude compatible power – a genuine order-of-magnitude bound, not a fit.

Numerology warning. Within $p = 2$, the coefficient ($\frac{1}{2}$) and the reference mass are *not* fixed by the single anchor. Different references land somewhere in the neutrino window (e.g. $\xi^2/2 \cdot m_e \approx 4.5$ meV, $\xi^2 \cdot m_e \approx 9$ meV, $\xi^2/2 \cdot \sqrt{m_e m_\mu} \approx 65$ meV). This is *not* evidence but the P35 trap: coefficient and reference need a derivation, not tuning. Computed: kinetik_xi_anteil.py.

.6 The structural lead: $m_{\text{window}} = \xi \cdot \omega$

The Doc. 047 velocity expression $v_v = c(1 - \xi^2/2)$ is exactly equivalent to a ξ^1 relation. For $m \ll p$ one has $v/c = 1 - \frac{1}{2}(m/p)^2$; substituting gives $(m/p)^2 = \xi^2$, hence

$$m_{\text{window}} = \xi \cdot p \quad (\approx \xi \cdot \omega \text{ via } \tilde{T} \cdot m = 1). \quad (1)$$

The " ξ^2 " in the notation "mass = $\xi^2/2 \cdot m_e$ " is thus presumably the *square image* of a fundamental ξ^1 relation: the static rest-mass is ξ times the kinetic scale. This is the natural anchor for a derivation *via the duality*, because $\omega = 1/\tilde{T}$ is exactly the duality quantity. The open question therefore shifts from "why ξ^2 ?" to "why ξ^1 between rest-mass and kinetic scale, and what sets ω absolutely?".

.7 What remains to be derived

Three points are explicitly open:

1. **The reference/scale – largely closed** (next section). The two lab anchors select the reference as the *single* FFGFT scale $E_0 = \sqrt{m_e m_\mu}$; in its units $\omega/E_0 = \xi/2$ and $m_{\text{window}}/E_0 = \xi^2/2$. Residual: the *one* structural statement $\omega = \xi E_0$ (exactly one fractal ξ -power).
2. **The coefficient.** The factor $\frac{1}{2}$ in $\xi^2/2$ is not derived; possibly it is purely the velocity square $(\frac{1}{2}(m/p)^2)$, i.e. no free factor at all.
3. **The spectrum.** A single ξ -fractal value sets the *scale*, not the *spectrum* (Δm_{sol}^2 , Δm_{atm}^2 , mixing angles). That demands the specified T^7 dynamics (the second open computation, Doc. 285).

.8 Closing the scale question: the lab anchors select E_0

The first open point – what sets ω absolutely – closes further than a free parameter would. In FFGFT there is only *one* absolute scale; everything else is dimensionless ratios. The question is therefore not "which number" but "which of the existing scales". With $m_{\text{window}} = \frac{\xi^2}{2} \cdot \text{reference}$, the two *lab* anchors (oscillation floor $\gtrsim 49$ meV; KATRIN < 0.45 eV) bracket the reference:

reference	m_{window}	floor $\gtrsim 49$ meV	KATRIN < 0.45 eV
m_e	4.5 meV	<i>fails</i> (too light)	yes
$E_0 = \sqrt{m_e m_\mu}$	65 meV	yes	yes
$M \approx 314$ MeV	2.8 eV	yes	<i>fails</i>
m_μ	940 meV	yes	<i>fails</i> (too heavy)

m_e is too light, M and m_μ too heavy – *only* $E_0 = \sqrt{m_e m_\mu}$ fits, and that is exactly the one natural FFGFT scale (Doc. 282, $\lambda_e \lambda_\mu = E_0$). The reference is thus not a free quantity but the existing scale, selected by the lab.

In units of E_0 everything then becomes a pure ξ -power times $\frac{1}{2}$:

$$\frac{\omega}{E_0} = \frac{\xi}{2}, \quad \frac{m_{\text{window}}}{E_0} = \frac{\xi^2}{2}. \quad (2)$$

No free parameter remains. The $\frac{1}{2}$ is the velocity square $(\frac{1}{2}(m/p)^2)$, Doc. 047) – kinematic, not a tuning factor; its *structural* origin is the AM sideband factor (see the signal-engineering reading below).

What honestly remains. The lab window (49–800 meV) is a factor $\sim$ few wide: it selects the *scale* E_0 sharply (excluding the alternatives by orders of magnitude) but does not pin the exact coefficient by itself. What remains is exactly *one* structural statement,

$$\omega = \xi E_0, \quad (3)$$

– E_0 suppressed by exactly *one* fractal ξ -power ($D_f = 3 - \xi$). This is no longer “free scale, self-referential” but a non-circular computation step: ω as the ξ -suppressed eigenvalue of the connection-Laplacian in the time-extended Hilbert structure (Doc. 283 §8). It closes the point – or falsifies it. Computed: `schliesse_omega.py`.

.9 The ξ^1 mechanism: lifting the zero mode by the dimension deficit

The remaining residual – why $\omega = \xi E_0$ with exactly *one* ξ -power – has a concrete mechanism. The connection-Laplacian on T^4 has an exact zero mode (the holonomy-flat constant). The fractal correction $g = \eta(1 + \xi f)$ acts as a diagonal potential perturbation and lifts it; first-order perturbation theory gives

$$\delta\lambda_0 = \xi \langle 0|f|0 \rangle = \xi \cdot \bar{f}, \quad (4)$$

i.e. the power ξ^1 *precisely when* the correction has a non-vanishing torus average $\bar{f} \neq 0$; for $\bar{f} = 0$ the first order vanishes and the mode lifts only at ξ^2 .

$D_f = 3 - \xi$ is a dimension *deficit* – a systematic, non-volume-preserving change of the measure –, hence $\bar{f} \neq 0$. So the power is *one*. Numerically (`xi1_eigenwert.py`): the zero-mode lift scales with $p = 0.999$ for a deficit ($\bar{f} \neq 0$) versus $p = 1.98$ for a volume-preserving ripple ($\bar{f} = 0$) – so the power is not accidental but follows from the nature of $D_f = 3 - \xi$.

Status. The ξ^1 power is thus *derived* (zero-mode lift by the dimension deficit), not asserted. What remains is only the $O(1)$ coefficient – kinematically the $\frac{1}{2}$ (Doc. 047) – and confirming that the physical kinetic state is this lifted zero mode. With that, $\omega = \frac{\xi}{2} E_0$ is closed up to an $O(1)$ factor – and that $O(1)$ factor is identified structurally, in the signal-engineering reading below, as the sideband $\frac{1}{2}$.

.10 The signal-engineering reading: AM sidebands and mode-locking

FFGFT grew out of communications engineering and the Fourier duality time $\leftrightarrow$ frequency (Doc. 001a); $\tilde{T} \cdot m = 1$ is that duality. In the dynamic sector the same language yields two concrete statements – one solid, one a candidate.

The $\frac{1}{2}$ is the AM sideband factor. The fractal correction $g = \eta(1 + \xi f)$ is formally an amplitude modulation of the carrier with modulation index ξ (Doc. 057: multiplicative = AM, additive = FM). Marcel’s kinetic prefactor $A(t) = 1 + \varepsilon(t)$, $|\varepsilon| \ll 1$, is the same object ($\xi \leftrightarrow \varepsilon$). For a carrier $\cos(\omega_c t)$ one obtains necessarily

$$\cos(\omega_c t) [1 + \xi \cos(\omega_m t)] = \cos(\omega_c t) + \frac{\xi}{2} [\cos((\omega_c + \omega_m)t) + \cos((\omega_c - \omega_m)t)], \quad (5)$$

so the sidebands sit at amplitude $\xi/2$. The kinetic scale $\omega = \frac{\xi}{2} E_0$ is therefore the *first AM sideband of the mass carrier* E_0 : the ξ from the modulation index (= dimension deficit,

previous section), the $\frac{1}{2}$ from the sideband split ($\cos \cdot \cos = \frac{1}{2}(\text{sum} + \text{difference})$). This unifies the ξ^1 power (lifted zero mode = first sideband) and the $O(1)$ coefficient into *one* mechanism: the $\frac{1}{2}$ is not inserted kinematically but is structural – the velocity square ($\frac{1}{2}(m/p)^2$, Doc. 047) is its kinematic shadow, not its origin.

$\theta = 2/9$ **as a mode-locking ratio (candidate).** The phase layer of the fork (Doc. 283) reads: *dynamic operator-phase* R (HLV, skew-adjoint, driven by $\dot{\phi}$) against *fixed scalar phase* $\theta = 2/9$ (FFGFT). Locking a dynamic phase onto a fixed rational value is well-defined in signal engineering: injection locking, i.e. Arnold tongues. A driven nonlinear oscillator locks onto rational rotation numbers p/q ; the stable plateaus form a devil's staircase. $\theta = 2/9$ would then be the lock plateau of a rational rotation number – which connects directly to FFGFT's *winding numbers* $(n_\theta, n_\phi, n_3, n_4)$ (Doc. 210): a locked rational winding. Mode-locking is a quantization rule – rational values are the stable fixed points – exactly the kind of “holonomy/fixed-point/quantization rule” the open θ point demands. The *open static* θ input would thereby acquire a *dynamic* origin in the same sector.

What stays a candidate. The sideband point is solid: it reproduces $\xi/2$ necessarily. The mode-locking reading is a *mechanism candidate*, not a derivation. There is a tension: small denominators ($1/2, 1/3, 2/5$) have the widest tongues, while $2/9$ (denominator 9) would be a narrower plateau; *why exactly* $2/9$ locks is open. But the reframing is productive: “why this value?” becomes “which Arnold tongue, and why is it stable?” – a testable question about a driven oscillator, not about a bare number. An honest bound on the candidate: the rotation number $2/9$ is *dimensionless* (cycles), the Koide offset $\theta = 2/9$ is a *radian* angle – $2\pi \cdot 2/9 = 4\pi/9 \neq 2/9$ –; reading the two as the same is an additional, open assumption, not part of the computed candidate (cf. Doc. 282).

Z_3 protection: the denominator is explained, the numerator is not. The FFGFT mass core is a Z_3 circulant (Doc. 282); if the driving nonlinearity carries this symmetry (invariant under $\theta \rightarrow \theta + 1/3$, i.e. only harmonics $k = 3, 6, 9, \dots$), the picture shifts. A structurally faithful Z_3 drive widens the tongues with 3-power denominators *selectively*: at equal criticality $1/3$ becomes 24× wider and the whole 9-family 11–12×, while the control tongue $2/5$ (denominator 5, not a multiple of 3) stays *flat* (0.97×) – the signature of genuine symmetry protection, not an artefact (`z3_modeLock_test.py`). This answers “why a denominator-9 lock at all”: the Z_3 structure of the core; $2/9$ is no longer absurdly narrow but symmetry-protected robust. *The numerator stays open*: the symmetry protects the 9-family jointly ($1/9 = 2/9 = 4/9$ identical), it does not single out $2/9$, and $1/3$ stays wider. The question thus splits cleanly – “why $2/9$ ” = “why denominator 9” (Z_3 , settled) + “why numerator 2 in the family” (open, exactly Marcel's BD17A holonomy domain).

The numerator needs a holonomy. The 9-family is a Z_3 orbit; a single 3-fold drive is therefore numerator-blind. The numerator is selected only by a *second* 3-structure with a definite relative phase ϕ – a holonomy that lives in the embedding space ($3'$ of the doubling $6 = 3 \oplus 3'$, $T^4 \rightarrow T^7$, Doc. 285), not in the core. With two 3-fold harmonics ($h=3, h=9$) and a relative phase ϕ (not gauge-removable) the degeneracy breaks (`z3_holonomie_test.py`, bisection): $1/9$ and $4/9$ stay paired, $2/9$ is the singlet that moves – for $\phi \in (\pi, 2\pi)$ $2/9$ is the *widest* tongue (clearest margin at $\phi = 3\pi/2$), crossover exactly at $\phi = \pi$. *Honestly*: the effect is small ($\sim 0.5\%$), ϕ is free (FFGFT does not fix it), and the minimal model splits only 2-against-1. But the structural statement stands: a holonomy *can* select $2/9$, and $2/9$ is exactly the member it acts on. The open question is thereby precisely shifted – no longer

"why 2/9" but "what locks ϕ into the band where 2/9 wins" – and that is Marcel's BD17A "locked holonomy".

One fractal correction, two orthogonal phases. Holonomy and time modulation spring from the *same* fractal correction $g = \eta(1 + \xi f)$ – the driving nonlinearity is f (Doc. 051, Doc. 270). One might fear that treating them as independent is inconsistent. The computation (`eine_fraktale_quelle.py`, 3-adic Weierstrass function) shows the opposite: they are two *orthogonal* functionals of f . The time phase $A = 2\pi\xi\langle f \rangle$ reads the *mean* $\langle f \rangle$ (the DC part, the dimension deficit, $O(\xi)$); the holonomy $\phi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$ reads the *phases of the 3-power harmonics* ($O(1)$). The two are analytically decoupled – $\langle f \rangle$ is independent of the harmonic phase (numerically to $2 \cdot 10^{-16}$), since $\cos(k\theta + \psi)$ has vanishing mean for every ψ . The worry is thereby resolved: neglecting A does not touch ϕ , they read different Fourier parts of the *same* f . The real consistency lives elsewhere: A and the ξ^1 mass lift *both* depend on $\langle f \rangle$ – so the time phase is tied to the mass window, not free. And no free lock: a phase-coherent fractal ($\psi = 0$) gives $\phi = 0$ (degeneracy), whereas 2/9 requires f 's 3-power harmonics to carry a relative phase in $(\pi, 2\pi)$ (a ramp $\delta = \pi/2$ per octave hits $\phi = 3\pi/2$ exactly, the 2/9 maximum – but as a choice, not a derivation). Thus "what locks ϕ " is concretely named: not some arbitrary phase, but the relative phase of the 3-power harmonics of the fractal correction f – the FFGFT-internal target for BD17A.

Demarcation: what is not a mechanism here. Modulation index and lock ratio are different quantities in different sectors: $\xi = 4/30000$ is the fixed AM index in the amplitude sector (it yields the sideband $\frac{1}{2}$), 2/9 is a rotation number in the phase sector. The lock ratio is *not* ξ . Any "derivation" that sets ξ to an $O(1)$ value (such as $1/3$ or $\tan(\pi/9) \approx 0.364$) in order to force 2/9 has left FFGFT – ξ is fixed and orders of magnitude smaller ($\sim 10^{-4}$). Likewise the open question is the *stability* of the 2/9 plateau, not an algebra between ξ and 2/9; and the memory kernel the locking would rest on is the discrete-spectral, oscillating kernel of Doc. 282 ($\sum_k g_k^2 \delta(\omega - \omega_k)$, with backflow/revivals), not a dispersive all-pass.

.11 Recursion as feedback: loop gain, regimes, discipline

FFGFT grew out of communications engineering (Doc. 001a), and its recursions are *feedback loops*. This places the results above in a frame – and fixes what about them is *free* and what is not.

The loop gain is a ξ quantity – and fixed. The magnitude of the loop gain decides the behaviour. In FFGFT it is geometrically given, not a dial:

- $|v| < 1$ (negative feedback): the loop contracts. The ξ -damping recursion $r(k+1) = r(k)(1 - \xi)$ (Doc. 275) has gain $1 - \xi$, only ξ below one – hence the enormous, slow scale tower; it runs to its only fixed point 0 and merely *passes through* $1/\varphi$ (no attractor).
- $|v| = 1$ (Barkhausen): marginal sustained oscillation.
- $|v| > 1$ (positive feedback): the loop runs away. The golden φ branch ($|v| > 1$) unrolls, the $-1/\varphi$ branch ($|-1/\varphi| < 1$) is the overdamped empty window (Doc. 285).

Decompactification is exactly this transition: *one* circle tips from the compact (contracting) into the runaway regime – the mass circle $\tilde{T} \cdot m = 1$ unrolls to non-compact time (Doc. 270), and that tipping *is* the measurement.

The phase selects itself. A feedback loop imposes no phase: the Barkhausen phase condition closes at the frequency the loop *finds*; in injection locking the lock phase slides within the lock range (Adler equation). Hence $\theta = 2/9$ is to be read as a *self-selected lock ratio* (rotation number), not as a tightly imposed phase. A "derivation" that prescribes a fixed phase $2/9$ and solves backwards for ξ mistakes this self-adjustment (cf. the demarcation above).

Two tiers. Linear and weak (ξ small): first order = the AM sideband $\frac{1}{2}$ (above, solid); a purely linear loop does not lock. Nonlinear: mode-locking onto rational rotation numbers = the $\theta = 2/9$ candidate (above); the lock-range width scales with the feedback strength ξ , and $\xi \sim 10^{-4}$ with denominator 9 makes the $2/9$ tongue extremely narrow – that is the open stability problem, not a missing piece of algebra.

Negative feedback as equalization. The fractal correction $g = \eta(1 + \xi f)$ (with K_{frak}/RG , Doc. 133/051) acts as broadband negative feedback that linearizes the bare carrier and sets $D_f = 3 - \xi$ – weak, with no sharp phase alignment required.

Discipline. This fixes how the tools are to be used *correctly*: (i) the loop gain (ξ , the eigenvalues $\varphi/-1/\varphi$) is fixed, never a dial – forcing a result by setting ξ is the cardinal error; (ii) the open questions are always *regime and lock range* (contracting/marginal/runaway; tongue width), never an algebra on the gain; (iii) the phase is self-selected, quantized values are lock ratios; (iv) the regime of the feedback carries the physics – negative feedback = compact stable scales + equalization, positive feedback = decompactification/measurement.

.12 Relation to the static sector

The static sector is closed and computed: the Z_3 mass operator ($r = \sqrt{2}$, $\theta = 2/9$) plus the mass circle yields Koide $2/3$, the lepton ratios and – as a pre-registered, matched-null-tested declaration – the BD12-XCORE interface (Doc. 285). The dynamic sector is the *complementary, open* side: the same parameter ξ , but as a *fraction* rather than an *eigenvalue*. A derivation of $m_{\text{window}} = \xi \cdot \omega$ would move the window-filling dynamics (Doc. 263, holographic scale projection) from description to computation, and the neutrino-like $3'$ candidate (Doc. 285) from conditional to derived.

Retrospect: two things both called " $2/9$ ". It is worth fixing the thread, for $\theta = 2/9$ appears in two roles. (i) $2/9$ as a *number/input*: the Z_3 circulant of the core uses it and reproduces the leptons (Koide), empirically closed – it was always there (Doc. 282; Doc. 285, the "protected-invariant" row $r = \sqrt{2}$, $Q = 2/3$, $\theta = 2/9$). (ii) $2/9$ as a *question* ("why this value", specifically the numerator): this becomes relevant only with the dimensional extension. The reason is structural, not a lack of computation: the bare Z_3 core is *numerator-blind* – $1/9, 2/9, 4/9$ are a Z_3 orbit, and a single compact 3-structure cannot in principle answer "why $2/9$ rather than $1/9$ ". The numerator is seen only by a second 3-structure with a relative phase = a holonomy = exactly what the extension provides ($3 \oplus 3', T^4 \hookrightarrow T^7$). The "why- $2/9$ " question thus has no home in the core; it is by nature an extension question.

Where this already stands in the corpus. Three documents already carry the thread. **Doc. 285** (dimension bridge) is the extension itself: $HLV \supset FFGFT, T^4 \hookrightarrow T^7$ ($7 = 4 + 3$, the three perp dimensions), the doubling $6 = 3 \oplus 3'$ ("literally"), the fork (FFGFT 3-fold

$\sqrt{2}/Q = 2/3/\theta = 2/9$ against HLV 5-fold $2/\sqrt{5}/\varphi$; $\theta = 2/9$ sits in the invariant row. **Doc. 283** (bridge and fork, §8) introduces $\theta = 2/9$ explicitly as a *complex holonomy*, the “operational consequence of compactness”, with the phase layer “dynamic operator-phase R (HLV) against fixed scalar phase $\theta = 2/9$ (FFGFT)” – there $2/9$ is already a holonomy/extension object, not a bare core number. **Doc. 282** is the core that *uses* $2/9$ (leptons), with the honest line “translation is not justification” – the notation *represents* $2/9$ but does not *ground* it.

The fixed point and the conclusion. Why the static sector is closed *at all*: the lepton masses sit at a *self-consistent fixed point* – the mass operator $\Phi_* = \sum_i m_i P_i$ is the non-trivial fixed point of the recursion $\mathcal{R}(\Phi_*) = \Phi_*$ (Doc. 203, lepton exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$); *that* is why it is exact, not a fitted value. From the lepton measurements one *reads off* this fixed point (that is the closure); fixed and open separate cleanly: $Q = 2/3$ is the operator condition $r = \sqrt{2}$ (algebraically fixed, Doc. 282), $\theta = 2/9$ is the holonomy (the numerator, open). Hence the conclusion: all the recent work above (feedback, mode-locking, holonomy test) is *no new arena* but the mechanistic, signal-engineering version of what $282 \rightarrow 283 \rightarrow 285$ already say structurally – $2/9$ is a holonomy, and a holonomy lives in the extension.

Magnitude and phase: why the split is inevitable. In signal-engineering language the whole finding collapses to a single, well-known theorem. The mass operator *is* a Z_3 circulant, and a circulant is diagonalized by the DFT – its eigenvalues are the DFT of its first row. The three lepton masses are thus literally a *three-point spectrum*: $\sqrt{2}$ is the *magnitude* of the spectral coefficients, $\theta = 2/9$ their *phase*. The radial/angular conclusion (Doc. 282) is therefore nothing but *magnitude spectrum versus phase spectrum*: everything determined sits in the magnitude (ξ as scale, the hierarchy as envelope, $Q = 2/3 = \sqrt{2}$ as amplitude ratio, $\langle f \rangle$ as DC term), everything open in the phase (θ , the holonomy $\varphi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$, the harmonic phases). The deep reason is the theorem itself: *a magnitude spectrum does not fix the phase spectrum* – same power, arbitrary shape. *That* is precisely why θ resists every FFGFT tool: the recursion $\mathcal{R}$ is purely real, hence pure gain, and real gain does not touch the phase. The orthogonality computed above *is* “magnitude $\perp$ phase”. From this the FFGFT $\leftrightarrow$ HLV division of labour even becomes a circuit diagram: FFGFT is the *real amplifier stage* (magnitude, hierarchy), HLV’s skew-adjoint R is the *all-pass/phase stage* ($e^{i\varphi}$, magnitude-preserving, rotating only the phase). A skew-adjoint operation *is* an all-pass. The missing angular principle is the phase stage – and the HLV side carries it, not FFGFT. (Limit of the same lens: mode-locking yields a *rotation number* (dimensionless), θ a *radian*; $2\pi \cdot 2/9 \neq 2/9$ – the lens identifies the *type* correctly, a locked phase, but the literal identification does not close.)

.13 Reproducibility

kinetik_xi_anteil.py (numpy-only): narrowing of the ξ -power against the bounds, the duality framing ($\tilde{T} = 1/\max(m, \omega)$), and the finding $p = 2$ with the explicitly open reference/-coefficient/spectrum gap. The script claims no derivation; it narrows. schliesse_omega.py: the lab selection of the reference E_0 (table) and the reduction $\omega/E_0 = \xi/2$, $m_{\text{window}}/E_0 = \xi^2/2$, with the ξ^1 remainder as the marked closing point. xi1_eigenwert.py: the zero-mode lift as ξ^1 from the dimension deficit ($p = 0.999$ vs $p = 1.98$). am_seitenband_modeLock.py: confirms the sideband ratio $\xi/2$ necessarily (deviation 0%) and measures the Arnold-tongue widths on the sine circle map – $2/9$ locks as a plateau but is $\sim 12\times$ narrower than $1/3$: candidate, not derivation. z3_modeLock_test.py: a Z_3 -symmetric drive (harmonics = multiples of 3) widens the 3-power-denominator tongues selectively ($1/3$ by $24\times$, the 9-family by $11\text{--}12\times$), while the control $2/5$ stays flat ($0.97\times$) – the denominator 9 is Z_3 -protected (Doc. 282), the numerator 2

stays open. `z3_holonomie_test.py`: two 3-fold harmonics ($h=3, h=9$) with a relative phase ϕ ; measured precisely (bisection), ϕ breaks the 9-degeneracy – $1/9, 4/9$ stay paired, $2/9$ moves and is the widest tongue for $\phi \in (\pi, 2\pi)$ ($\sim 0.5\%$, maximum at $\phi = 3\pi/2$); ϕ free $\Rightarrow$ shifts the open question to the ϕ -lock (BD17A). `eine_fraktale_quelle.py`: shows computationally (3-adic Weierstrass f) that the time phase and the holonomy are *orthogonal* functionals of the same fractal correction – $A = 2\pi\xi\langle f \rangle$ reads the mean ($O(\xi)$), $\phi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$ the harmonic phases ($O(1)$), $\langle f \rangle$ phase-independent to $2 \cdot 10^{-16}$; the time phase is tied to the ξ^1 mass lift (both $\langle f \rangle$), the holonomy phase stays an independent datum (coherent $\Rightarrow \phi = 0$; $\delta = \pi/2$ ramp $\Rightarrow \phi = 3\pi/2$).